

Template for Computer Programming I Practical Recording

Date	:	10/04/2025
Problem Specification	:	To count votes for two candidates across multiple precincts, determine who won each precinct, compute the total votes, and calculate each candidate's winning percentage.
Assumption	:	Does not validate for negative votes. Assumes input is always correct and in expected format. No input error handling if user gives invalid characters.
Limitation	:	Only two candidates: Polly and Ernest. The number of votes entered per precinct is accurate and non-negative. The user indicates whether more precincts need to be entered.
Input	:	Number of votes for Polly and Ernest in each precinct. User input (Y/N) to indicate more precincts.
Processing	:	Tally votes per candidate, count who won each precinct, and compute vote percentage.
Output	:	Total percentage of votes each candidate received. Number of precincts each candidate won. Number of ties.
Algorithm	:	Step1: Start Step2: Create a function as getvotePrecinct() 1.Declare a int array voteinpresint[] with a size of 2. 2.prompt "number of votes for Polly in each precinct" 3.get input from user for Polly ‘s Votes assign for voteinpresint[0] 4. Prompt "number of votes for Ernest in each precinct" 5.get input from user for Ernest’s Votes and assign for voteinpresint[1] 6.return value voteinpresint Step3: declare variables as votesForPolly, votesForErnest, totalPolly, totalErnest, response, percentagePolly, percentageErnest, totalVotes Step4: Prompt "Election Day Vote Counting Program" Step5: Initialize pollyWoncount ← 0 ernestcountWon ← 0 tiescount ← 0 totalPolly ← 0 totalErnest ← 0

Step6: Create an array a[] for store the return values from Step

And assign value for votesForPolly ← a[0];

votesForErnest ← a[1];

Step7:compute

totalPolly ← totalPolly+votesForPolly;

totalErnest ← totalErnest+votesForErnest;

Step8: Prompt "the user whether or not there are more precincts to report "

Step9:get input y or n from user and assign for response

Step10: if(a[0]>a[1]) true

pollyWoncount ← pollyWoncount+1

else if(a[0]<a[1]) true

ernestcountWon ← ernestcountWon+1

else

tiescount tiescount+1

Step11:repeat step 5 to step 10 while (response == "Y")

Step12: compute

totalVotes ← totalPolly+totalErnest

percentagePolly ← (totalPolly/totalVotes)*100

percentageErnest ← (totalErnest/totalVotes)*100

Step13:print percentagePolly, percentageErnest

Step14:print pollyWoncount , ernestcountWon, tiescount

Step15: End the program

Programme listing

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Programme file **attached**

Test data and expected output

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1. Test data: Polly: 30, Ernest: 20

Polly: 25, Ernest: 35

Polly: 50, Ernest: 50

User input: Y, Y, N, N

Expected output: Polly percentage: 50.0000

Ernest percentage: 50.0000

Polly won count: 1

Ernest won count: 1

Ties count: 1

Output obtained for test data : 1. Test data: Polly: 30, Ernest: 20
Polly: 25, Ernest: 35
Polly: 50, Ernest: 50
User input: Y, Y, N, N
Obtained output: Polly percentage: 50.0000
Ernest percentage: 50.0000
Polly won count: 1
Ernest won count: 1
Ties count: 1

Discussion : Proper use of array to fetch multiple inputs from one method.

Conclusion : This program calculates the vote count for two candidates over multiple precincts, helping determine the winner and distribution of votes through modular methods and loop control.